ASM5 REPORT

For this project, in addition to writing some assembly, you also need to write a short report detailing some features of your code. To find the required information, load up your solution (asm5.s) and a test that calls the function subsCipher. The easiest way to do this is to put both files in the same directory (with no other .s files) and then turn on the "Compile all files in the directory" option of MARS.Assemble these two files, and do a quick test run to ensure your code works.For this report, you need to examine the arrays str and map in memory.Set a breakpoint at the JAL subsCipher instruction and step into the function. Fill out the values on registers $a0 and $a1 and write their meaning for this function.

REGISTER	VALUE WHEN CALLING subsCipher	What is the value?
$a0	0x1001001e	Address of the input string
$a1	0x1001009c	Address of the map array

Using the function arguments, examine the memory on MARS (.data) and take a (full screen)1 screenshot of the data on arrays str and map. Include these screenshots in your report. Complete the following table:

	Address of element	Value of element
The first element of the array of str	0x1001001e	Asdf - First element
The last element of the array of str	0x10010029	
The first element of the array of map	0x1001009c	
The last element of the array of map		

Data Segment

Address	Value (+0)	Value (+4)	Value (+8)	Value (+c)	Value (+10)	Value (+14)	Value (+18)	Value (+1c)
0x10010000	- - - -	- - - -	- - - -	- - - -	o < \0 \n	r e h t	\0 : >	s a \0 \n
0x10010020	1 \0 f d	4 \n 3 2	7 \n 6 5	T \0 9 8	q e h	k c i u	o r b	f n w
0x10010040	j x o	e p m u	v o d	t r e	l e h	y z a	. g o d	e w q \0
0x10010060	u y t r	a p o i	g f d s	l k j h	v c x z	\0 m n b	D C B A	B A F E
0x10010080	F E D C	D C B A	B A F E	F E D C	z Y x \0	x Z y X	\0 y y x	. . . .
0x100100a0	. . . .	. . . .	. . . .	. . . .	. . . .	. . . .	. . . .	. . . .
0x100100c0	. . . .	. . . .	. . . .	r e w q	i u y t	. . p o	. . . .	d s a .
0x100100e0	j h g f	x z l k	n b v c	B N M m	Z X C V	. J K L	. . . .	F G H .
0x10010100	P A S D	Y U I O	W E R T	8 9 0 Q	4 5 6 7	. 1 2 3	. . . .	. . . \0
0x10010120	. . . .	. . . .	. . . .	. . . .	. . . .	. . . .	. . . .	. . . .
0x10010140	. . . .	. . . .	. . . .	@ !) .	^ % $ #	. (* &	. . . .	b a . .
0x10010160	f e d c	j i h g	n m l k	r q p o	v u t s	z y x w	. . . .	B A . .
0x10010180	F E D C	J I H G	N M L K	R Q P O	V U T S	Z Y X W	. . . .	+ \n \0 .
0x100101a0	- - - -	- - - -	- - - -	- - - -	- - - -	- - - -	- - - -	- - - -
0x100101c0	- - - -	- - - -	- - - -	- - - -	- - - -	- - - -	+ - - -	| \n
0x100101e0	a M	c i g	u l a V	p (e	e p p o	r f d	t m o	s e h

C:\Users\reric\OneDrive\Desktop\CSC252\asm5\asm5.s - MARS 4.5

File Edit Run Settings Tools Help

Run speed at max (no interaction)

Edit Execute

Text Segment

Bkpt	Address	Code	Basic	Source
	0x004004a8	0x27bd0018	addiu $29,$29,0x000...	217: addiu $sp, $sp, 24
	0x004004ac	0x03e00008	jr $31	218: jr $ra
	0x004004b0	0x27bdffe8	addiu $29,$29,0xfff...	225: addiu $sp, $sp, -24
	0x004004b4	0xafbe0000	sw $30,0x00000000($...	226: sw $fp, 0($sp)
	0x004004b8	0xafbf0004	sw $31,0x00000004($...	227: sw $ra, 4($sp)
	0x004004bc	0xafa40008	sw $4,0x00000008($29)	228: sw $a0, 8($sp) # save arg1 for later
	0x004004c0	0xafa5000c	sw $5,0x0000000c($29)	229: sw $a1, 12($sp) # save arg2 for later
	0x004004c4	0x27be0014	addiu $30,$29,0x000...	230: addiu $fp, $sp, 20
✓	0x004004c8	0x0c10006c	jal 0x004001b0	233: jal subsCipher

Data Segment

Address	Value (+0)	Value (+4)	Value (+8)	Value (+c)	Value (+10)	Value (+14)	Value (+18)	Value (+1c)
0x10010000	- - - -	- - - -	- - - -	- - - -	o < \0 \n	r e h t	\0 : >	s a \0 \n
0x10010020	1 \0 f d	4 \n 3 2	7 \n 6 5	T \0 9 8	q e h	k c i u	o r b	f n w
0x10010040	j x o	e p m u	v o d	t r e	l e h	y z a	. g o d	e w q \0
0x10010060	u y t r	a p o i	g f d s	l k j h	v c x z	\0 m n b	D C B A	B A F E
0x10010080	F E D C	D C B A	B A F E	F E D C	z Y x \0	x Z y X	\0 y y x	. . . .
0x100100a0	. . . .	. . . .	. . . .	. . . .	. . . .	. . . .	. . . .	. . . .
0x100100c0	. . . .	. . . .	. . . .	r e w q	i u y t	. . p o	. . . .	d s a .

0x10010000 (.data) Hexadecimal Addresses Hexadecimal Values ASCII

Coproc 1 Coproc 0

Registers

Name	Number	Value
$at	1	0x10010000
$v0	2	0x00000000
$v1	3	0x00000000
$a0	4	0x1001001e
$a1	5	0x1001009c
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x31c6d22b
$t1	9	0x00000000
$t2	10	0x00000000
$t3	11	0x00000000
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0xac634282
$s1	17	0x9b5fd006
$s2	18	0x13737c9d
$s3	19	0x923d0494
$s4	20	0x715c203d
$s5	21	0xbf5be260
$s6	22	0xcd057c10
$s7	23	0x93618268
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffefe0
$fp	30	0x7fffeff4
$ra	31	0x004002c8

Mars Messages Run I/O

10:52 PM 4/9/2025